standout project. A dream project of Chief Minister Nitish Kumar, it included a four-lane road across the Ganga connecting north and south Bihar. Raised on a concrete frame across five sections, the bridge was to connect the Kosi region in the northeast, comprising the districts of Khagaria, Saharsa, Madhepura and Supaul, with the southeast districts of Bhagalpur, Munger and Jamui. The bridge contract was signed in November 2014, with Singla Construction qualifying as the lowest bidder among six candidates. The total cost was set at Rs 1,710.8 crore with the bridge component coming in at Rs 859 crore. The rest of the money was to be used for land acquisition.

The slow pace of the project was already a concern for the Bihar Rajya Pul Nirman Nigam Ltd (BRPNNL), the government agency monitoring the project (it was to be ready by November 2019). Government sources say Singla was granted three extensions but puts the blame on delays in land acquisition for the approach roads to the bridge. However, But months before the last deadline was to expire (June 2022), the Pylon 5 disaster happened. The BRPNNL then stopped the work and approached IIT Roorkee to investigate the collapse.

THE COLLAPSE OF THE UNDER-CONSTRUCTION BRIDGE IS A SETBACK FOR CM NITISH'S DREAM PROJECT THAT AIMS TO CONNECT PARTS OF NORTH AND SOUTH BIHAR

The investigation report, prepared by Umesh Sharma and Sanjay Chikarmane (both IIT Roorkee), and Subodh Jain, retired member (engineering) of the Indian Railways, came as an eye-opener. It identified a "partial failure of the match casting process that led to compressive stress concentration". Simply put, the precast segments that were supposed to fit perfectly and transfer the load across the entire surface of the bridge failed to do so. The report cited images to suggest that the wet stitch (which maintains aggregate interlock and minimises movement of concrete slabs at the cracks) failed to form a bond and "delaminated" from the concrete surface. "The philosophy of creating a good joint is that the base material should fail before the joint. This has not happened at the wet stitch locations," says the report, which added that NDT (non-destructing test on concrete) of the wet stitches indicates a loss of integrity at "some of the tested wet stitch locations". The NDT is conducted to find the compressive strength and other properties of concrete from existing structures. Officials believe the flaws that led to the collapse of Pylon 5 are responsible for the second crash too. On both occasions, the extradosed module had reportedly collapsed.

Divided in five sections (1 to 5), the bridge is being built on the reinforced cement concrete (RCC) method that involves segmental construction. Overall, there are 29 pylons under the bridge, involving 672 pre-cast segments. Out of these, the first collapse resulted in the loss of 54 segments whereas the second, on June 4, saw 108 segments going under. INDIA TODAY sent S.P. Singla Construction a questionnaire seeking their response on the accidents, but they did not respond.

The BRPNNL was clearly alarmed after the April 2022 collapse. Government records show that at a review meeting on December 14, 2022, the department rejected the argument of S.P. Singla's design and technical consultant, Canadian firm McElhanney, that the hairline cracks developed after the April 2022 accident were not structural in nature and would not affect the safety of the bridge. BRPNNL put it on record that the fallen segments may have impacted the stress cables. It also mentioned that the hairline cracks in the pylon at pier 5 extended throughout the circumference and cannot be ignored. On January 17, road construction department secretary Sandeep K.R. Pudakalkatti signed an order asking for the removal of the remaining segments on pier 5 and its reconstruction, with the contractor bearing the cost.

Even as these segments were being removed in March 2023, the IIT Roorkee team was told to investigate the entire bridge and submit a final report. Now the June 4 collapse has raised a huge question mark on the project itself. Departmental sources suggest that one of the points being considered is if there was a design defect (as the IIT-Roorkee report suggests) that makes the superstructure unviable. One consolation for the Bihar government is that the bridge contract was scripted with Singla responsible for the EPC (engineering, procurement and construction), so the risk responsibilities rest with the contractor. The contract also has a defect liability period of five years, which means the contractor is legally required to repair any defects that appear in the period after project's completion. There's also the Rs 100 crore in bank guarantees that the state can seize if the project goes south. Hearing a PIL on the bridge collapse on June 14, the Patna High Court asked Singla Construction to submit details of the project report and purchase receipts of material used in the construction. It also asked the government for an action taken report.

Bihar deputy CM Tejashwi Yadav, who also holds charge of the road construction department (RCD), has admitted to having doubts about the bridge design after the IIT Roorkee report. As he indicated, the government might order the demolition of the existing structure to rebuild it from scratch, which will not be good news for Singla. For the RCD, a major concern is again fixing responsibility. The government has now suspended BRPNNL executive engineer in charge of the project, Yogendra Kumar, and also axed MD Niraj Saxena. But is it too little, too late? Bihar has a legacy of bridges collapsing, especially in the past few years. Perhaps it is time for a system-wide check on how many more are in peril. ■